

1. Listed option

premium, payoff,
settlement, Greeks



2. Cashflow map

NAV return,
conditional premium



3. Joint risk model

$$\Sigma_O = [B \ I] \Sigma_{f, \varepsilon} [B \ I]^\top$$



4. R1 allocation

net mean - variance
CVaR, stress, margin, cash



5. Validation gates

integer feasibility,
walk-forward, ruin gate